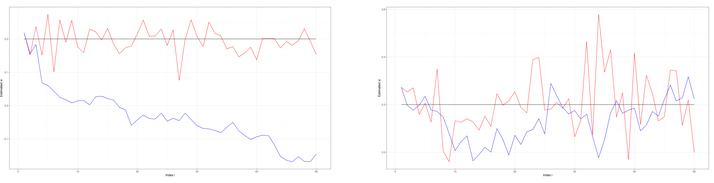
Multiscale data disaggregation with application to climate studies

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Motivation and background

- Two methods to estimate the regular variation index $1/\xi$: the block maxima method and the POT method.
- The POT method is often preferred but requires choosing a threshold u below which data are not used.
- **Problem:** How to pick u?
- Things get even more complicated when aggregating data:



(a) Threshold at 90%. (b) Threshold at 99%.

Figure 1. Estimates of ξ for $S_d = \sum_{k=1}^{d} X_k$, *vs.* X_d, $d = 1, \ldots, 50$.

- Like [5], one could make ξ vary with d, the number of aggregated variables:

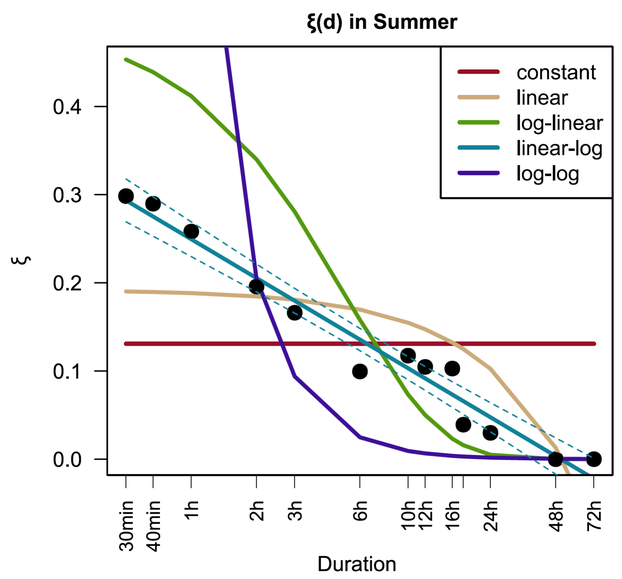


Figure 2. Dependence of ξ on duration in summer at a station in Zurich (taken from [5])

- However, that approach contradicts *Breiman's theorem*: ξ is stable under convolution.
- This leads to discrepancies between theory and the models adopted.

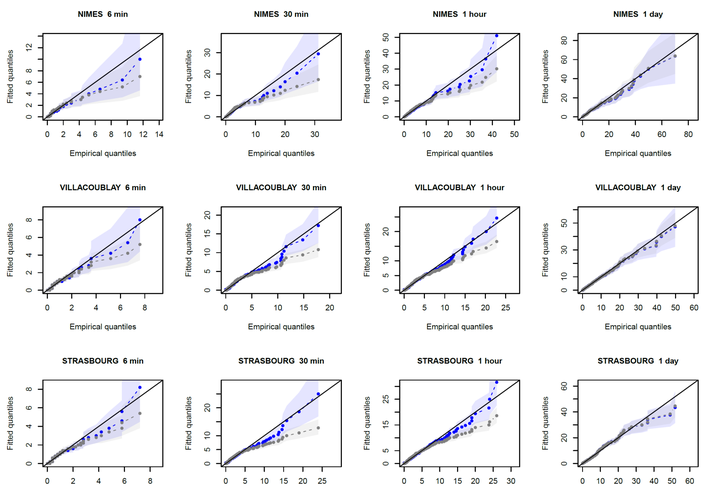


Figure 3. Discrepancies between the theory and the model (taken from [2])

- Predictions based on aggregated values will tend to underestimate the intensity of precipitations at a lower time scale.

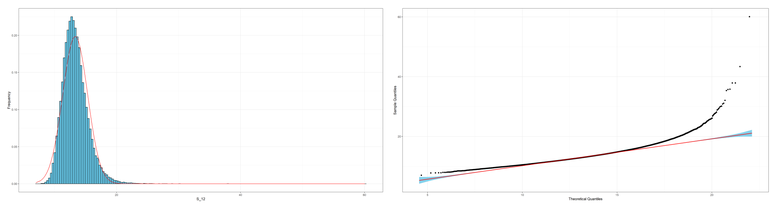
The cause of the problem

- Depending on how big the threshold u is compared to d, the distribution beyond u of the aggregated data is roughly either Pareto or Gaussian:

$$\mathcal{L}(S_d \,|\, S_d \geq u) \simeq \begin{cases} \mathcal{N}_u(d\mu, d\sigma^2) & \text{if } d \gg u \\ u\mathcal{VP}(\xi^{-1}) & \text{if } u \gg d, \end{cases}$$

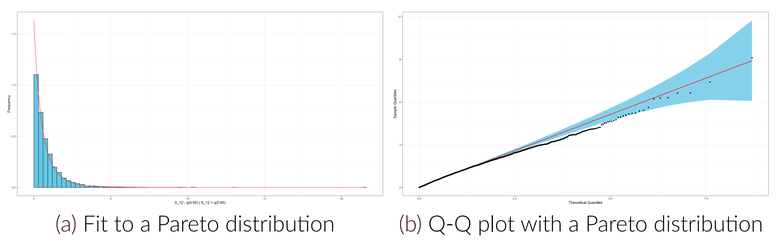
  where $\mathcal{N}_u(d\mu, d\sigma^2)$ denotes the Gaussian distribution conditional on taking values greater than u and $\mathcal{VP}(\xi^{-1})$ the Pareto distribution on $[1, +\infty)$.
- The coarser the time scale is, the higher u must be to observe a convergence to a Pareto distribution.
- Otherwise, the central limit theorem kicks in and makes $\mathcal{L}(S_d \,|S_d \geq u)$ approximately normal (over $[u, +\infty)$).

Comparison to conditional Gaussian and Pareto distributions



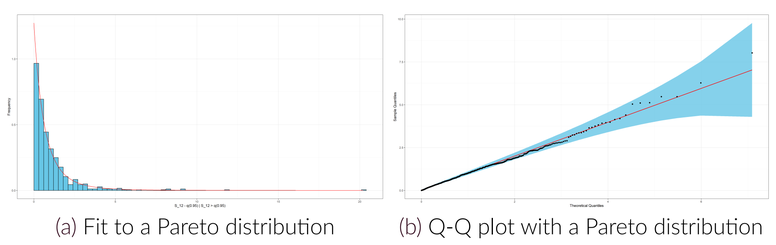
(a) Fit to a normal distribution (b) Q-Q plot with a normal distribution

Figure 4. Non-truncated ($u = 1$) normalized samples of S_{12} *vs.* normal distribution.



(a) Fit to a Pareto distribution (b) Q-Q plot with a Pareto distribution

Figure 5. Truncated ($u = \hat{q}_n(0.95)$) normalized samples of $\mathcal{L}(S_{12} - u \,|S_{12} \geq u)$ *vs.* Pareto distribution.



(a) Fit to a Pareto distribution (b) Q-Q plot with a Pareto distribution

Figure 6. Truncated ($u = \hat{q}_n(0.99)$) normalized samples of $\mathcal{L}(S_{12} - u \,|S_{12} \geq u)$ *vs.* Pareto distribution.

Our approach

- Estimating the distance from $\mathcal{L}(S_d \,|\, S_d \geq u)$ to a Pareto or conditional Gaussian distribution provides insight about how big u should be.
- In this approach, the time scale d is fixed and the threshold is a function $u = u(d)$ of the time scale.

Probability metrics

Common examples of metrics between probability distributions include the *Kolmogorov metric*:

$$\mathrm{d}_K(\mu, \nu) = \sup_{x \in \mathbf{R}} |\mu[x, +\infty) - \nu[x, +\infty)|$$

and the *Wasserstein metric* associated with d:

$$\mathrm{d}_W(\mu, \nu) = \inf_{\substack{X \sim \mu \\ Y \sim \nu}} \mathbb{E}[d(X, Y)],$$

where d is a metric on $\mathbf{R}$, and the infimum is taken over every coupling (X, Y) of μ and ν. For instance, $d(x, y) = |x - y|$ on $\mathbf{R}$, or $d(x, y) = |\log x - \log y|$ on $(0, +\infty)$.

The second-order regular variation assumption

- **Definition:** $\overline{F}$ is *second-order regularly varying* with indices $-1/\xi$ and $\rho \leq 0$ (noted $\overline{F} \in 2\mathrm{RV}_{-1/\xi,\rho}$) if there exists $A : \mathbb{R}_+^* \to \mathbb{R}$ vanishing at infinity such that

$$\frac{\overline{F}(tx)}{\overline{F}(t)} \underset{t\to\infty}{=} x^{-1/\xi} + x^{-1/\xi}\left(\frac{x^\rho - 1}{\rho}\right) A(t) + o\big(A(t)\big), \quad x > 0.$$

- **Theorem [1]:** The *auxiliary function* A of X determines the speed of convergence of $\mathcal{L}(U(t)^{-1}X \,|\, X \geq U(t))$ to the Pareto law:

$$d_W\Big(\mathcal{L}\big(U(t)^{-1}X | X \geq U(t)\big), Z\Big) \underset{t\to\infty}{\sim} CA(t),$$

  where Z has the Pareto distribution of parameter $1/\xi$ and

$$U(t) = F^{-1}(1 - t^{-1}).$$

- Important case: i.i.d. data with generalized Pareto distribution (see [7, 4]).

The extended generalized Pareto distribution (egpd)

The distribution function of a unit egpd is

$$F_{B,\xi^{-1},\kappa}(x) \coloneqq B\big(H_{\xi^{-1}}(x)^\kappa\big) = B\Big(\big(1 - (1 + \xi x)^{-\frac{1}{\xi}}\big)^\kappa\Big)$$

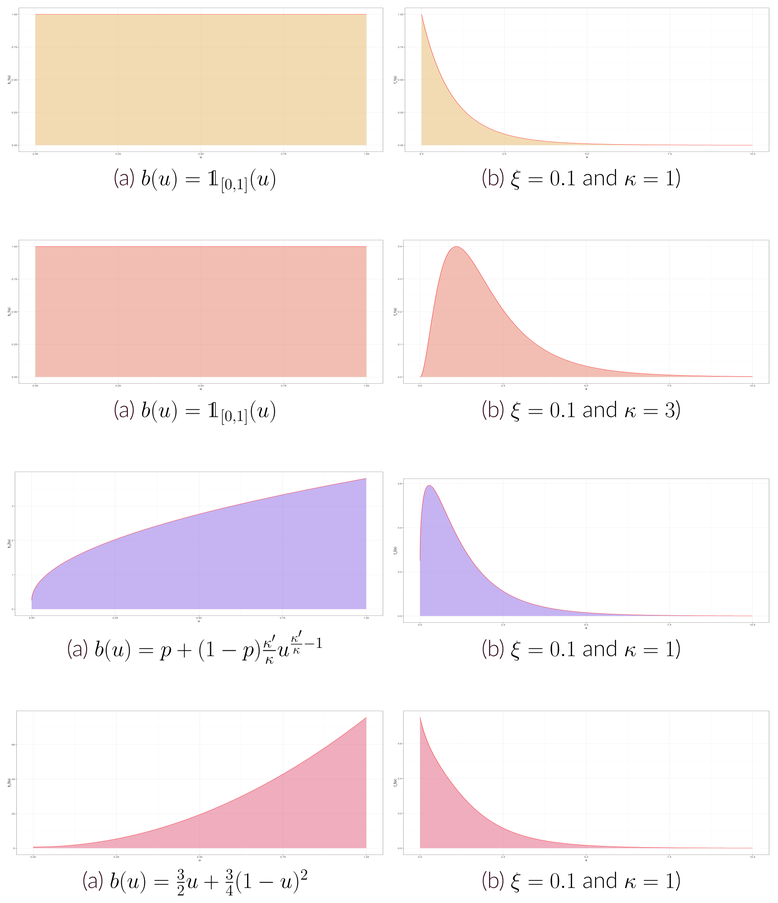
where

- $1/\xi > 0$ models the large values of the distribution
- $\kappa > 0$ plays a similar role, but for the small values
- $B : [0, 1] \to [0, 1]$ is an absolutely continuous cdf with density b, such that:

$$\lim_{t\to 0^+} \frac{B(t)}{t} = b(0) \quad \text{and} \quad \lim_{t\to 0^+} \frac{B(t) - 1}{t - 1} = b(1)$$

  are finite and positive.

Examples



(a) $b(u) = \mathbb{1}_{[0,1]}(u)$ (b) $\xi = 0.1$ and $\kappa = 1$)

(a) $b(u) = \mathbb{1}_{[0,1]}(u)$ (b) $\xi = 0.1$ and $\kappa = 3$)

(a) $b(u) = p + (1-p)\frac{\kappa'}{\kappa}u^{\frac{\kappa'}{\kappa}-1}$ (b) $\xi = 0.1$ and $\kappa = 1$)

(a) $b(u) = \frac{3}{2}u + \frac{3}{4}(1-u)^2$ (b) $\xi = 0.1$ and $\kappa = 1$)

Figure 10. On the left: the density b of B. On the right: the density of the corresponding egpd

- $B(u) = u$
- $B(u) = pu + (1-p)u^{\kappa'/\kappa}$, where $p \in [0, 1]$
- $B(u) = 1 - (1 + \delta^{-1})(1 - u^\kappa)\Big(1 - \frac{(1-u^\kappa)^\delta}{1+\delta}\Big)$, where $\delta > 0$
- $B(u^\kappa) = \sum_{i=1}^{m} \omega_{m,i} \int_0^{u^\kappa} \beta_{i,m-i+1}(y)\, \mathrm{d}y$, with $\sum_{i=1}^{m} \omega_{m,i} = 1$.

Main results

- **Proposition [3]:** The following implication holds:

$$t \mapsto 1 - B(1 - t^{-1}) \in 2\mathrm{RV}_{-1,\rho} \implies \overline{F}_{B,\xi^{-1},\kappa} \in 2\mathrm{RV}_{-1/\xi,\tilde{\rho}}$$

  for a certain explicit $\tilde{\rho}$. The corresponding auxiliary function can be made explicit as well.
- **Proposition [3]:** The following equivalent holds:

$$d_W\Big(\mathcal{L}\big(U_d(t)^{-1}S_d \,|\, S_d \geq U(t)\big), Z\Big) \underset{t\to\infty}{\sim} CA_d(t),$$

  with C independent of d, $U_d(t) = F_{S_d}^{-1}(1 - t^{-1})$ and A_d the auxiliary function of S_d (see [6]).
- $A_d(t)$ is either independent of d or behaves like $dA(t)$, where A is the auxiliary function of the X_i.

Outlooks and future work

- Similar results when the target distribution is Gaussian remain to prove; the following bound constitutes a first step in that direction:

$$d_{\mathrm{K}}\big(\mathbb{P}_{X \,|\, X \geq u}, \mathbb{P}_{Y \,|\, Y \geq u}\big) \leq 2 \min\left(\frac{1}{\mathbb{P}(X \geq u)}, \frac{1}{\mathbb{P}(Y \geq u)}\right) d_{\mathrm{K}}\big(\mathbb{P}_X, \mathbb{P}_Y\big).$$

- Designing a systematic approach to deduce an optimal threshold u from the previous asymptotic bounds.
- Applying the same line of reasoning to the small values of a sum of egpd random variables (focusing on κ instead of ξ).
- Generalizing to time series (temporal dependence).
- Generalizing to higher dimension (spatial dependence).

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